

$$P(B_i|A) = \frac{P(A|B_i) \times P(B_i)}{\sum_{i=1}^n P(A|B_i) P(B_i)}$$

$$P(A|B) = P(A/B) \times P(B)$$

$$P(A_1 \cap A_2 | B) = P(A_1|B) \times P(A_2|B) \quad \{\because \text{Ind. } A_1, A_2\}$$

$$P(C) = P(X \in C)$$

$$E(g(x)) = \int_{-\infty}^{\infty} g(x) f(x) dx$$

$$P(X+Y=k) = \sum_{i=0}^k P(X=i) P(Y=k-i)$$

Jensen's Inequality
 $E(g(x)) \geq g(E(x))$: Convex

$$E(cX) = cE(X)$$

$$E(X+Y) = E(X) + E(Y)$$

$$E(aX+b) = aE(X) + b$$

$$E(aX+bY) = aE(X) + bE(Y)$$

$$E(AX) = AE(X)$$

$$E(U^T X) = U^T E(X)$$

$$V(aX+b) = a^2 V(X)$$

$$V(aX+bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{Cov}(X,Y)$$

$$V(AX) = A \Sigma A^T$$

$$\text{Cov}(U^T X, V^T Y) = U^T \Sigma V$$

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y) \\ = E((X-E(X))(Y-E(Y)))$$

$$\text{Corr}(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{V(X)} \sqrt{V(Y)}}$$

$$\text{Cov}(aX+b, cY+d) = ac \text{Cov}(X,Y)$$

$$E(g(X,Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y) dx dy$$

$\Sigma \rightarrow$ Constraint

$\rightarrow$ Positive semi definite $\rightarrow A \Sigma A^T$

$\rightarrow$ Positive definite for normal

Multivariate Normal $\rightarrow f_X(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu))$

$\rightarrow$ All marginals / conditionals are Normal

$\rightarrow$ All combinations (AX) will also be Multivariate Normal

if $A \Sigma A^T$ is PSD.

Law of Total Probability

$$P(A|C) = \sum_i P(A|C, B_i) P(B_i|C)$$

$$E(X|C) = \sum_i E(X|C, B_i) P(B_i|C)$$

$$f_{P(X)} = \frac{f_{P_i, X}}{f_X} = \frac{f_{X|P} \times f_P}{\int_{-\infty}^{\infty} f_{X|P} f_P(v) dv}$$

Geometric $\rightarrow$ no. of trials before 1st success

Poisson $\rightarrow$ no. of times event 1 occurs in unit time

$$X+Y \rightarrow \text{Poisson}(\lambda_1 + \lambda_2)$$

Uniform $\rightarrow F(x) \rightarrow \frac{x-a}{b-a} \quad (a < x < b)$ | Use area to calculate $P(X < x)$

Pareto $\rightarrow$ tail modeling $(x \geq \lambda) \quad | \quad f(x) = \frac{\alpha}{x^{\alpha+1}}$

Exponential $\rightarrow$ memory less | $F(x) = 1 - e^{-\lambda x}$
 "Time remaining for an event" $x > 0$

$$C(X) = E\varphi(B_C)$$

$$\sum_{i=1}^n x_i = Ga(n, \beta) \text{ (Independent)}$$

Gamma $\rightarrow$ Chi-square special case of Gamma
 waiting time for K events (K, λ)

Normal $\rightarrow$ linear combination of Independent N is also Normal

Indicator $f^n \rightarrow$ used for Expectation, $E(X) = P(X)$

can be added without being Independent

$$P(X,Y \in A) = \iint_A f_{X,Y}(x,y) dy$$

Marginal $\rightarrow$ Integrate out others

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

$$\text{Conditional } f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

[Range of f^2 of Y (Area)]

$\rightarrow$ Positive definite for normal

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$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy$$

$$E(X) = \int E(X|B_i) P(B_i)$$

$$E(Y) = \int_{-\infty}^{\infty} E(Y|X=x) f_X(x) dx \\ = E(E(Y|X))$$

Independence

$$f_{X,Y}(\mathbf{x}, \mathbf{y}) = f_X(\mathbf{x}) f_Y(\mathbf{y})$$

$$E(h_1(X)h_2(Y)) = E(h_1(X))E(h_2(Y))$$

Implies 0 Correlation / 0 Covariance & Normal / Not Ind. (X) Trivial on field $\{\Omega, \phi\}$

Multivariate + 0 Covariance Normal

Iterated Conditioning

$$E(X) = E(E(X|Y))$$

$$V(X) = V(E(X|Y)) + E(V(X|Y))$$

$$E(X|Y) \neq E(X|Y=y)$$

$$E(X|Y=y) = E(X|Y=Y(y))$$

Moments Generation Function

$$M_X(t) = E(e^{tX})$$

$$E(X^n) = \frac{\partial^n}{\partial t^n} M_X(t) \Big|_{t=0}$$

$$M_Y(t) = \prod_{i=1}^n M_{X_i}(t)$$

Sum of RV Independent

Convergence $M_{X_n} \rightarrow M_X \Rightarrow X_n \xrightarrow{D} X$ as $n \rightarrow \infty$

Measure Moment

σ field $\rightarrow \phi \in \mathcal{F}$

$A \in \mathcal{F} \quad A^c \in \mathcal{F}$

$\infty A_1, A_2, \dots \in \mathcal{F} \quad \cup_i A_i \in \mathcal{F}$

$E(E(X|F)/G) = E(X/F)$ if $F \in G$

$E(E(X/G)/F) \rightarrow$

$E(Y/X) = E(Y/\sigma(X))$

$X \rightarrow F$ measurable then $\sigma(X) \in \mathcal{F}$

$E(X|F) = X \mid E(X|F) = X E(Y|F)$

$E(X+d|F) = (E(X|F) + d)$ if ϕ convex

$E(\phi(X|F)) \geq \phi(E(X|F))$

Central Limit Theorem

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{D} Z \sim N(0,1)$$

$\sigma^2 > 0$ (defined)

$\bar{X}_n \sim N(\mu, \sigma^2/n)$

$\Sigma X_i \sim N(n\mu, n\sigma^2)$

$\bar{X} \rightarrow$ sample mean

$S^2 \rightarrow$ sample variance

$\Sigma_{i=1}^n (X_i - \bar{X})^2$

Delta Method

If $\frac{Y_n - \mu}{\sigma_n} \xrightarrow{D} Z$ and $\sigma_n \rightarrow 0$ if $n \rightarrow \infty$, $g \rightarrow$ Diff at μ & $g'(\mu) \neq 0$

$\frac{g(Y_n) - g(\mu)}{\sigma_n} \xrightarrow{D} Z$

$|g'(\mu)| \sigma_n$ or $g(Y_n) \sim N(g(\mu), (g'(\mu))^2 \sigma_n^2)$

Chi-Squared Distribution (K d.o.f)

$X \sim \text{Gamma}(K/2, 1/2)$

$Z \rightarrow N(0,1)$

$E(X) = K$

$V(X) = 2K$

$X = \sum_{i=1}^K Z_i^2 \rightarrow \text{IID}$

$\hookrightarrow$ chi with K d.o.f

heavy tail $\leftarrow$ Student + distribution

$T = \frac{Z}{\sqrt{X/V}}$

$Z \rightarrow N(0,1)$

$V = \frac{X}{V-2}$ ($V \geq 3$)

$Y \rightarrow \text{Chi square}(V)$

$T(1) = \text{Cauchy}$ (E(T) doesn't exist if tail is fat)

Classic Results

$X_1, X_2, \dots$ IID normal (Independent & Identical Distribution)

① $\bar{X}$ and $S^2 \rightarrow$ Independent

② $(\bar{X} - \mu)/(\sigma/\sqrt{n}) \rightarrow N(0,1)$

$E(S^2) = \sigma^2$

$S^2 \text{ of } \sigma^2 = \frac{\sqrt{2} \sigma^2}{\sqrt{n-1}}$

$(n-1)S^2/\sigma^2 \rightarrow \text{Chi square}(n-1)$

$(\bar{X} - \mu)/S/\sqrt{n} \rightarrow T_{dist}(n-1)$